

so $[\phi]$ is a class in $\mathcal{F}^k H_{-1}(\text{gr}_{\mathcal{L}}^i \mathfrak{h})$. We now consider the maps

$$\mathcal{F}^k H_{-1}(\text{gr}_{\mathcal{L}}^i \mathfrak{h}) \rightarrow \mathcal{F}^k H_{-1}\left(\frac{\mathfrak{h}}{\mathcal{L}^{i+1} \mathfrak{h}}\right) \rightarrow \mathcal{F}^k H_{-1}\left(\frac{\mathfrak{h}}{\mathcal{L}^{i+1} \mathfrak{h} + \mathcal{F}^{k+1} \mathfrak{h}}\right) \xleftarrow{f} \mathcal{F}^k H_{-1}\left(\frac{\mathfrak{h}}{\mathcal{L}^{i+1} \mathcal{F}^k \mathfrak{h} + \mathcal{F}^{k+1} \mathfrak{h}}\right)$$

The last group is where the k -th intermediate gauge triviality class ϑ_k^i lives. But the image of this class in $\mathcal{F}^k H_{-1}(\mathfrak{h}/\mathcal{L}^{i+1} \mathfrak{h} + \mathcal{F}^{k+1} \mathfrak{h})$ is equal to the image of $[\phi] \in \mathcal{F}^k H_{-1}(\text{gr}_{\mathcal{L}}^i \mathfrak{h})$, which vanishes by assumption. Therefore, to show that all classes ϑ_k^i vanish, it is sufficient to show that the map f is injective. Suppose we have $[\alpha] \in \ker(f)$. Then we can find λ such that

$$d^\psi \lambda \equiv \alpha \pmod{\mathcal{L}^{i+1} \mathfrak{h} + \mathcal{F}^{k+1} \mathfrak{h}}.$$

Let us set $\lambda' = \lambda^{(k-1)} + \lambda^{(k)} + \dots$. Since d^ψ is homogeneous of weight one, we must have

$$d^\psi(\lambda') \equiv \alpha \pmod{\mathcal{L}^{i+1} \mathfrak{h} + \mathcal{F}^{k+1} \mathfrak{h}},$$

since α is in $\mathcal{F}^k \mathfrak{h}$. Thus λ' is a primitive of α in $\mathfrak{h}/\mathcal{L}^{i+1} \mathcal{F}^k \mathfrak{h} + \mathcal{F}^{k+1} \mathfrak{h}$. $\square$

2.3. Application to formality of properadic algebras. The aim of the present article is to study the (intrinsic) coformality properties of Definition 1.33. To do so, we will use the obstruction theories developed in Section 2.1. Beforehand, let us recall the approach of formality as a deformation problem. In all this section, the ring R is a $\mathbb{Q}$ -algebra. Let $\mathcal{C}$ be a reduced weight-graded dg coproperad, e.g. $\mathcal{C} = \mathcal{Y}^{(n)\mathfrak{i}}$. Given any chain complex (A, d_A) , we have a convolution dg Lie admissible algebra

$$\mathfrak{g}_A = (\text{Hom}_{\mathbb{S}}(\overline{\mathcal{C}}, \text{End}_A), \partial, \star),$$

whose Maurer–Cartan elements are in bijection with $\Omega\mathcal{C}$ -structures on A . Recall that an ∞ -morphism between two $\Omega\mathcal{C}$ -algebra structures is an ∞ -isotopy if its first component is the identity. We denote by Γ_A the set of all ∞ -isotopies. The existence of gauge equivalences between Maurer–Cartan elements in $\mathfrak{g}_A$ corresponds to existence of ∞ -isotopies between the corresponding $\Omega\mathcal{C}$ -structures thanks to the following theorem.

Theorem 2.14 ([3, Theorem 2.16]). *If R is a $\mathbb{Q}$ -algebra, the set of all the ∞ -isotopies between $\Omega\mathcal{C}$ -algebra structures forms a group which is isomorphic through the graph exponential/logarithm maps to the gauge group of $\mathfrak{g}_A$*

$$\exp : ((\mathfrak{g}_A)_0, \text{BCH}, 0) \cong (\Gamma_A, \odot, 1) : \log.$$

Suppose that $\mathcal{C}$ is a reduced weight-graded coproperad (with no differential) and let H be a graded R -module. The associated convolution dg Lie admissible algebra $\mathfrak{g}_H$ is weight-graded Lie algebra in the sense of Assumptions 1, with the weight grading coming from that of $\mathcal{C}$. It also has an extra filtration where $\mathcal{L}^i \mathfrak{g}_A$ is all the operations with $(i+1)$ or more inputs. More precisely, the $(\mathbb{S}^{op} \times \mathbb{S})$ -module $\mathcal{C}$ has a direct sum decomposition under a *second* grading

$$\mathcal{C} = I \oplus \mathcal{C}_{(1)} \oplus \mathcal{C}_{(2)} \oplus \mathcal{C}_{(3)} \oplus \dots$$

where $\mathcal{C}_{(i)}$ is spanned by the operations with i outputs. This gives a weight decomposition

$$\mathfrak{g}_A = (\mathfrak{g}_A)_{(1)} \times (\mathfrak{g}_A)_{(2)} \times (\mathfrak{g}_A)_{(3)} \times \dots$$

and we can then define the extra filtration by

$$\mathcal{L}^i \mathfrak{g}_A = \prod_{j \geq i+1} (\mathfrak{g}_A)_{(j)}.$$

Assuming $\mathcal{L}$ is relatively bounded, that is bounded with respect to $\mathcal{F}$, we can apply all the methods of Section 2.1, giving the following applications of Theorem 2.10 and Theorem 2.13.